SMU Lyle School of Engineering



Reinforcement Learning

Planning and Learning with Tabular Methods

Sutton/Barto* Chapter 8

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018



Online Material

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Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
  - Dynamic programming
  - Monte Carlo methods
  - Temporal-difference learning
  - Multi-step bootstrapping
  - **Planning and learning with tabular methods**
- Part II: Approximate Solution Methods
  - Prediction and Control using Approximation
  - Eligibility Traces
  - Policy Gradient Methods
- Part III: Modern RL Methods
  - Deep Reinforcement Learning
  - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\text{argmax}_\text{a}\, f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s, a)$	expected immediate reward from state s after action a.
$r(s, a, s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Temporal Difference Learning

U_t	target for estimate at time t
δ_t	TD error

Planning vs. Learning

Planning vs. Learning

Model-based RL

- Requires knowledge of the MDP model.
- Methods: Dynamic Programming, heuristic search
- Use the known model to **plan**

Model-free RL

- Model is unknown.
- Methods: MC, TD
- **Explore + Learn** from interactions with the environment (sampling).



All methods do the following:

- **Generalized Policy Iteration:** Prediction and policy improvement.
- **One- or n-step lookahead** to future events (rewards, states, actions) and compute a backed-up value to update an approximate value function.

What is a Model

Model of the environment: Anything the agent can use to predict the outcome of actions (e.g., next state and reward).

Models can be:

- **Distribution models**: Specify the complete distribution of all possible outcomes.
  - For finite MDPs this is the function $p(s', r \mid s, a)$
  - Used by dynamic programming methods for planning.

- **Sample models**: Simulate the real environment by producing one outcome for an action. The outcome is a valid sample from the distribution model.
  - Can be used for MC and TD methods for learning. This is also called **trajectory sampling**.
  - Simulations are typically easier to create and can be used instead of slow, expensive, or dangerous real environments.

Note:

- The real environment can be seen as a physically represented model with sequential sampling access.

Can we integrate learning and planning?

Dyna

Integrating Planning, Acting, and Learning

Integrating Planning, Acting, and Learning

Direct RL is model-free Learning:

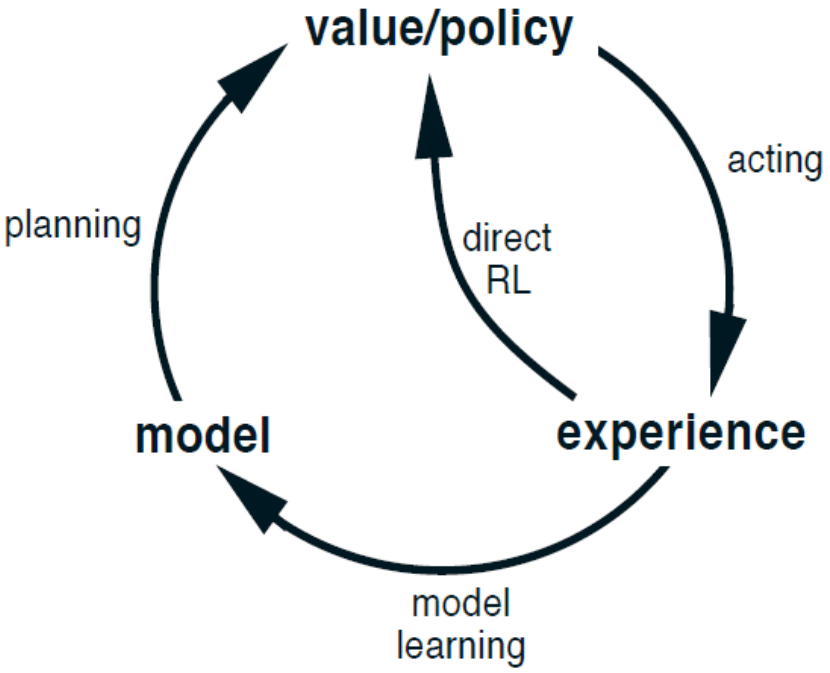
- Use sampled experience directly to **improve the value function/policy**. E.g., via one-step Q-learning.

Model Learning and Planning:

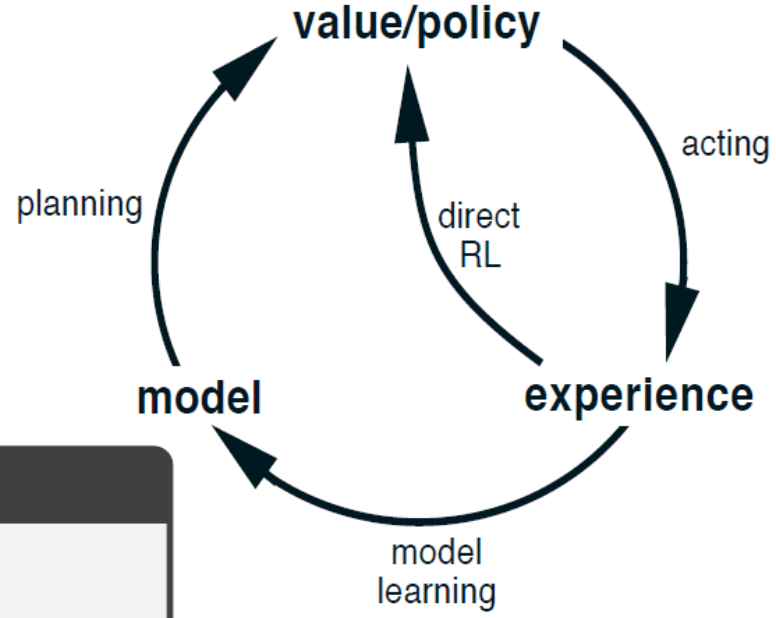
- Real experience is also used to **learn a model**. E.g., by recording each observed transition $S_t, A_t \rightarrow R_{t+1}, S_{t+1}$ to estimate $\hat{p}(S_{t+1}, R_{t+1} | S_t, A_t)$
- The model can be used for **planning** to improve the value function/policy. E.g., by sampling from the model to update the Q-values.

Advantage: Model learning and planning makes the approach much more sample efficient.

- Direct RL explores and makes sure we get good data.
- The model makes sure that we use experience is not just once.



Alg. Tabular Dyna-Q



Tabular Dyna-Q

Initialize $Q(s,a)$ and $Model(s,a)$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}(s)$

Loop forever:

(a) $S \leftarrow$ current (nonterminal) state

(b) $A \leftarrow \varepsilon$-greedy(S,Q)

(c) Take action A; observe resultant reward, R, and state, S'

(d) $Q(S,A) \leftarrow Q(S,A) + \alpha\left[R + \gamma \max_a Q(S',a) - Q(S,A)\right]$

(e) $Model(S,A) \leftarrow R, S'$ (assuming deterministic environment)

(f) Loop repeat n times:

$S \leftarrow$ random previously observed state

$A \leftarrow$ random action previously taken in S

$R, S' \leftarrow Model(S,A)$

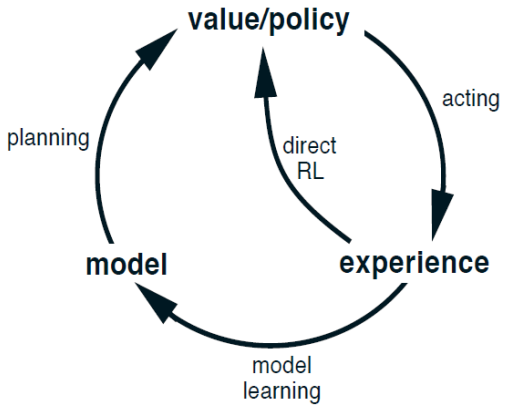
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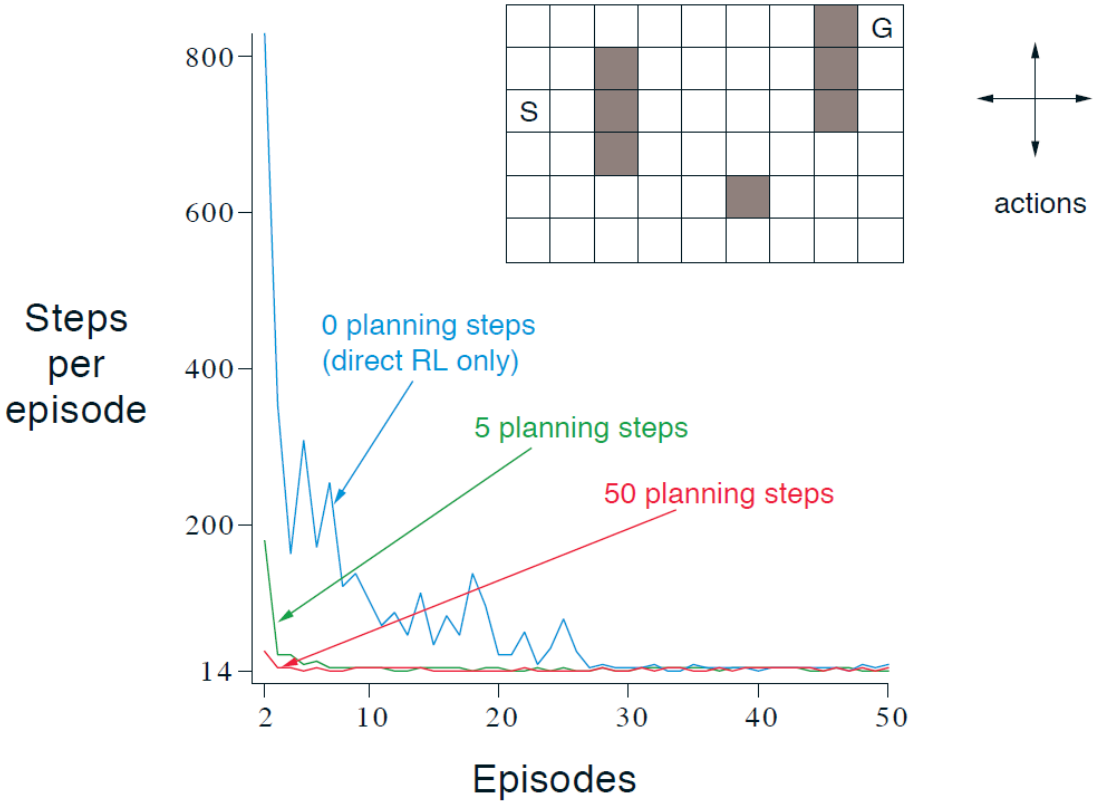
Direct RL: Q-Learning

Model learning: learn $p(s', r|s, a)$

Improve Q using the model (Planning)

The Effect of Planning





- **Benefit**: Adding planning speeds learning by reusing sampled experience.
- **Issues**: Planning is a problem if the model is wrong:
  - Based on an insufficient number of samples.
  - p-function approximation is not great.
  - Environment changes.
- Luckily, Dyna quickly corrects overly optimistic predictions since they will be selected for the next real action in the direct RL step.
- For changing environments:
  - Keep exploring.
  - Keep track of how out-of-date estimates are. I.e., for how long each action/state combination was not tried.

Code example: Learning and Planning with Dyna

See Code on https://michael.hahsler.net/Introduction_to_Reinforcement_Learning/

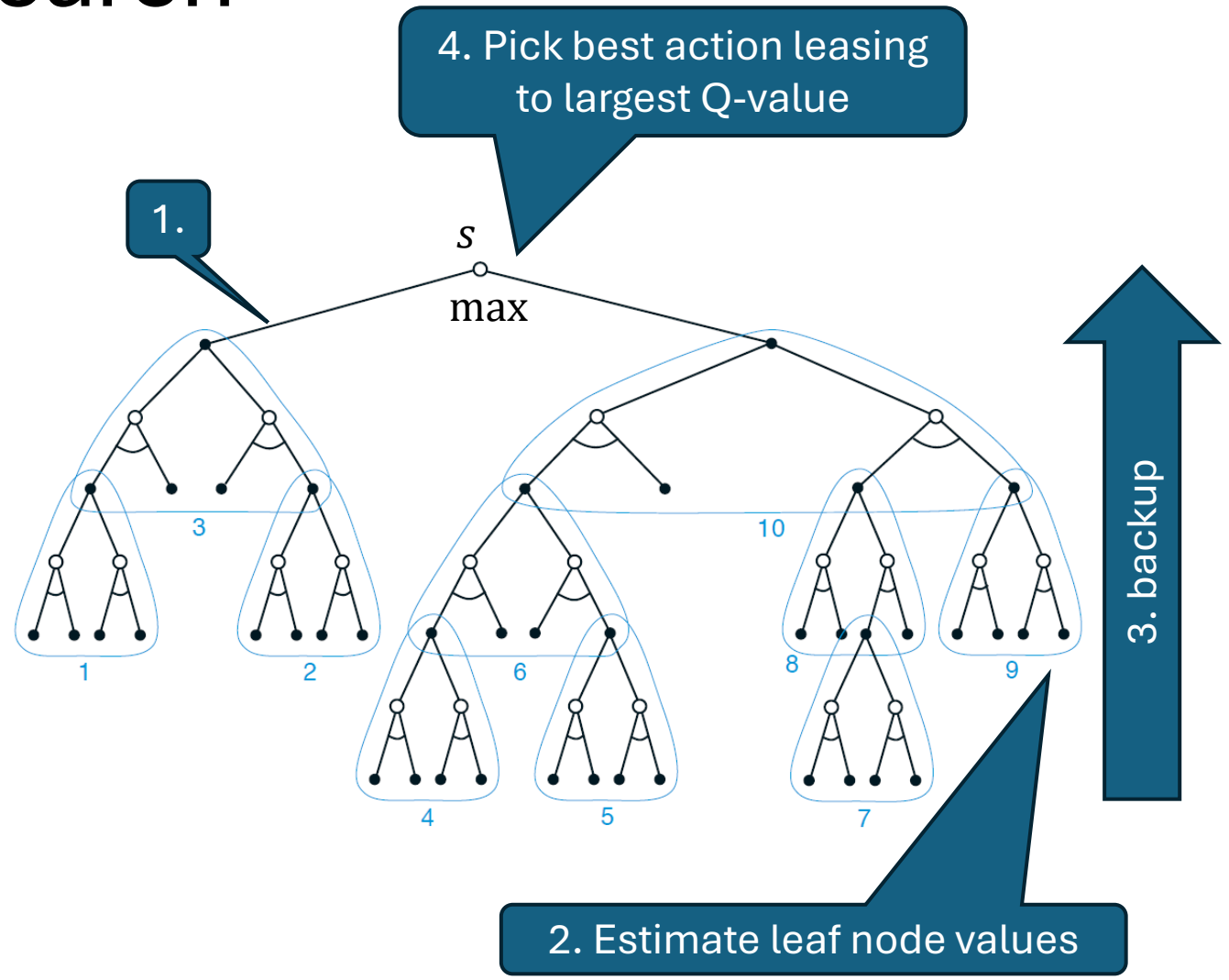
Heuristic Search and Rollout Algorithms

Decision-time planning

Option 1: Heuristic Search

At decision time in state s:

1. Consider a large tree of possible continuations (trajectories of a given length) for the current state.
2. Apply an approx. value function to the states in leaf nodes.
3. Back values towards the current state (using max). Similar to n-step estimation, but it always uses the best and not the observed action.
4. Choose the best next action for the current state.



Note: This is like heuristic cut-off min-max search for games (without min steps).

Option 2: Rollout Algorithms

Start with an initial policy π called the base policy.

At decision time in state s:

1. **Monte Carlo Policy Estimation**: Use many simulated trajectories to approximate all state/action values $\hat{q}_\pi(s,a)$ for the current state s.
2. **Choose the best action** $a = \underset{a}{\operatorname{argmax}}\, \hat{q}_\pi(s,a)$ according to the estimates. This is called the **rollout policy**.

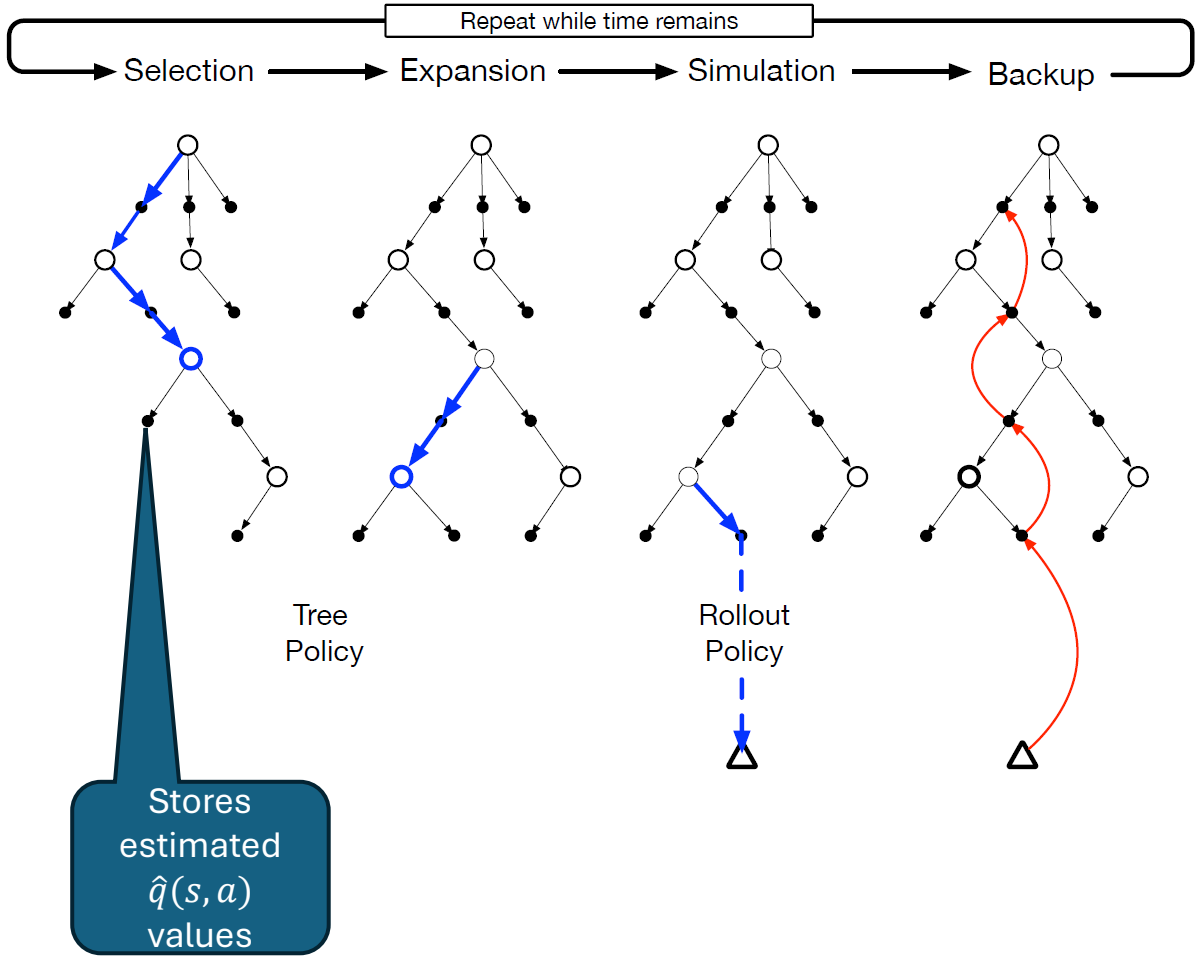
Properties:

- If the base policy is reasonable (i.e., somewhat similar to the optimal policy), rollouts are locally optimal.
- Rollout policy is guaranteed to be at least as good as the base policy (in expectation).
- **You get policy improvement without solving Bellman equations.**

Rollout Algorithm: Monte Carlo Tree Search

- MCTS is a rollout algorithm that stores some intermediate estimates in a depth-limited tree structure.
  - **Selection**: Pick the currently best leaf node (e.g., greedy, UCB).
  - **Expansion**: Add new nodes, but keep the tree small enough.
  - **Simulation**: From a selected node, simulate complete episode(s).
  - **Backup**: update the q-values in the tree with the result of the playout.

- **Advantage**: selection steers the simulation towards better trajectories. This avoids approximating the full value function.



MTCS can be seen as an RL method that selectively approximates q-values useful for choosing good actions.



What you Should Know



- We can have different models of the environment.
  - A full **distribution model**
  - A **sample model**

  - DP (planning) requires a distribution model because it uses expected updates.
  - Model-free RL, like MC and TD, can use a sample model to perform sample updates.
  - Decision-time planning can also use a sample model. E.g., rollout algorithms sample trajectories.

- Planning and learning are related and try to estimate the same value function.
  - **Learning**: Direct RL directly learns a value function
  - **Planning**: We learn a model used to estimate the value function.

Summary of Part I: Tabular Solution Methods

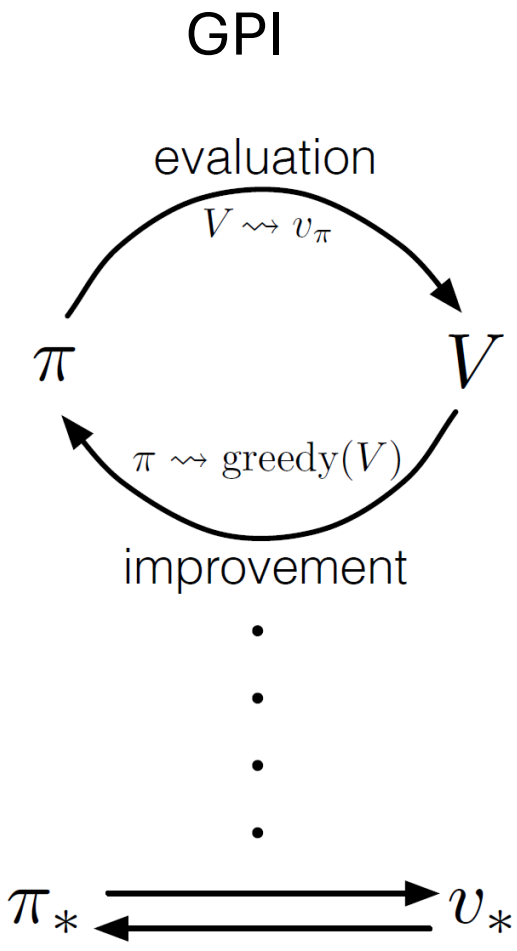
Common Ideas

1. **Policy Evaluation**: Estimate the value function by updates based on **backing up** values along actual or possible state trajectories.

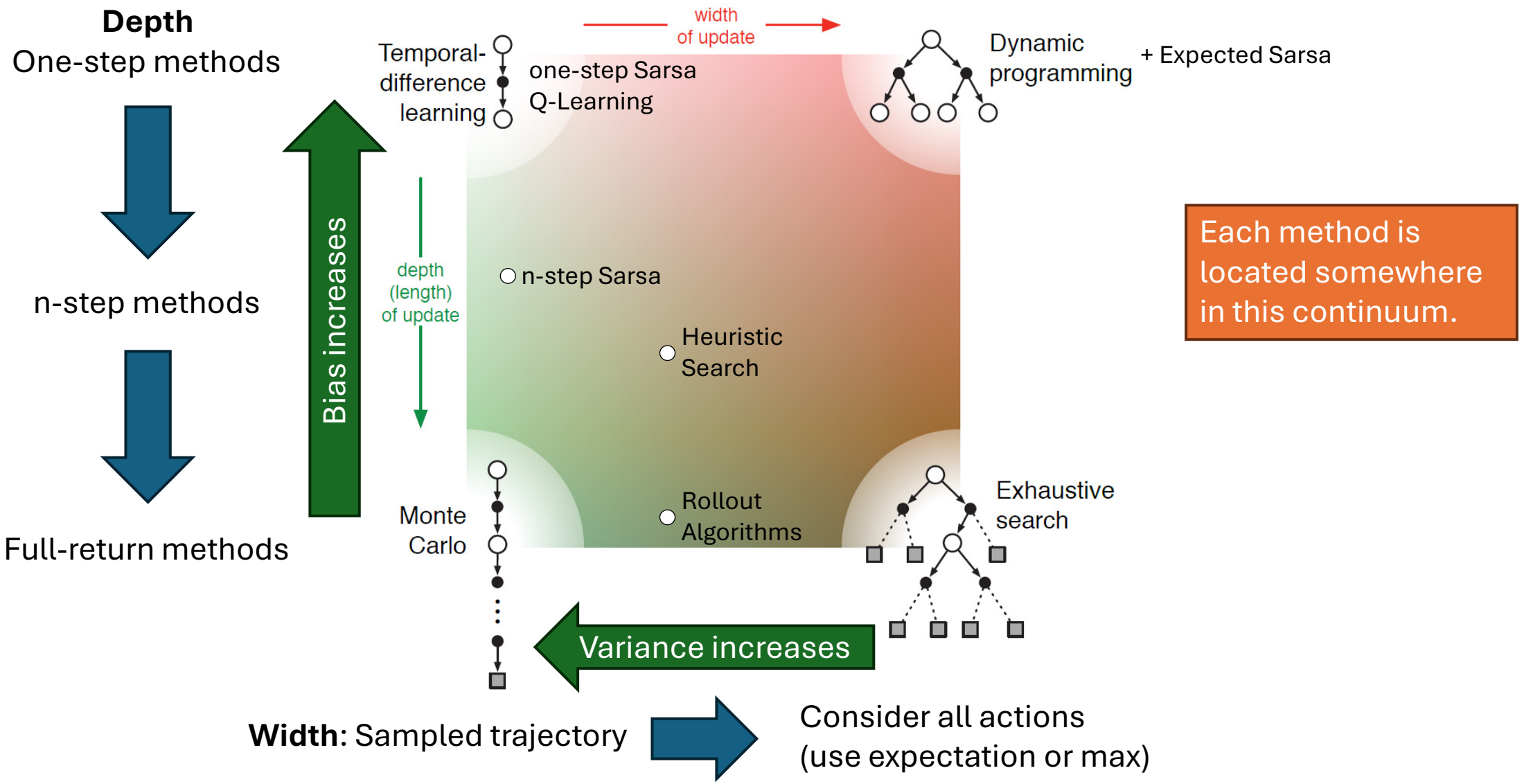
2. **Generalized policy iteration (GPI):** maintain an approx. value function and an approx. policy. Each is used to update the other. Converges due to the policy improvement theorem.

Convergence: Policy improvement theorem for greedy action choice.

Differences: How to perform policy evaluation.



Policy Evaluation: Update Depth vs. Update Width



On-Policy vs. Off-Policy

On-Policy: Evaluate or learn a policy that is used to sample data.

Off-Policy: Evaluate or learn a target policy different from the behavior policy.

Off-Policy is useful because:

- **Learn optimal policies** while using exploring behavior.
- **Reuse of data is possible**: offline, and batch RL on existing training data leads to better sample efficiency.

Off-Policy learning is harder because:

- **Requires distributional correction**: E.g., importance sampling which is not able to use all data when the behavioral policy is very different from the target policy.
- May suffer from **high variance** caused by importance sampling.
- **Instability with non-linear function approximation:** This is a major issue for deep RL

Other Dimensions

- **Definition of return**:
  - Episodic vs. continuing task
  - Discounted vs. undiscounted

- **Action values q vs. state values v**

- **Exploration vs. exploitation**: This is a trade-off. Many methods need exploration (covering behavior policies) for convergence. This is related to on s. off-policy learning.

- **Synchronous vs. asynchronous updates**: Are all states updated at once or one by one in some order?

- **Real vs. simulated experience**: Are real experiences, simulations, or both used for updates?

Relationship With Reflex Agents

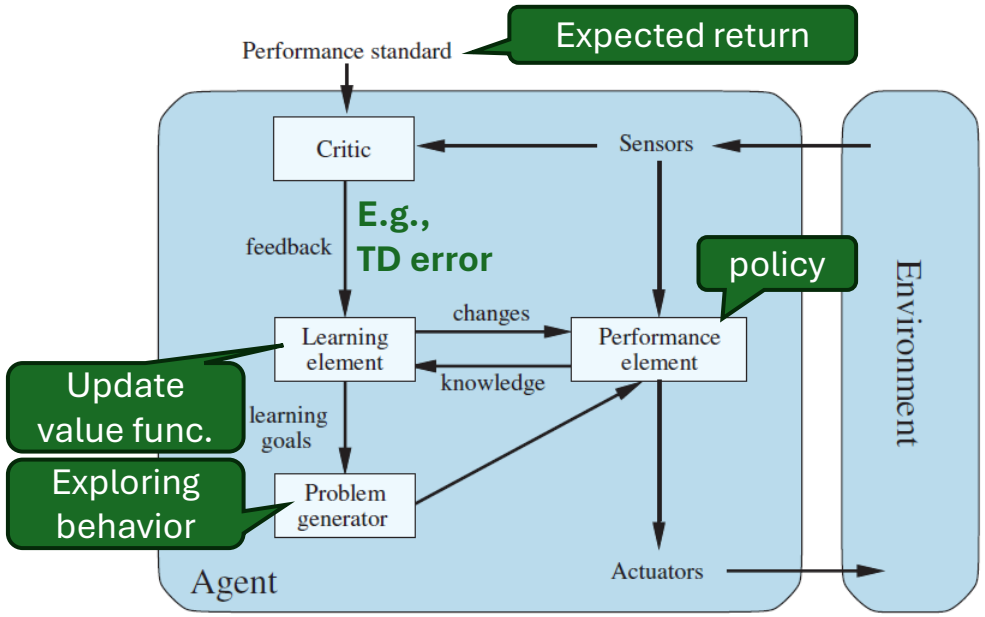
- Tabular methods create a policy that defines the action for each state.
- We can implement this policy by a set of rules, one for each state.
- The agent is **learning** because it improves the policy over time.
- For episodic tasks, tabular RL implements a **learning reflex agent.**
- For continuous tasks, where the rewards express utility, we get a **learning utility-based agent**.

Cases:

a) **Fully observable case:**
- DP uses the MDP for **planning** to create the optimal simple reflex agent.
- Model-free RL **learns** a simple reflex agent.

b) **Partially observable case**: The agent constantly updates its agent state to approximate the environment state. The agent stores the agent state and uses it to decide on actions. It learns a **model-based reflex agent.**

Model-free RL as A Learning Agent



Source: Artificial Intelligence: A Modern Approach, Chapter 2

Issues with Tabular Methods

1. Tables need a lot of **space**. A complete table with Q-values has size
$$O\left(|\mathcal{S}| \times |\mathcal{A}|\right)$$
The state space is typically extremely large, or even infinite, in continuous spaces.

2. Two states may be very similar and therefore should have similar Q-values. Tabular methods **cannot generalize across states** since each table entry is independent. We **need a lot of data** to estimate all entries.

In the next part of this course, we will talk about **approximate solution methods** to deal with these issues.